

DW03_lfsr_dcnto

LFSR Counter with Dynamic Count-to Flag

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Features and Benefits

- Dynamically programmable count-to value that indicates when the counter reaches a specified value
- High speed, area-efficient
- Asynchronous reset
- Terminal count

Revision History

Description

DW03_lfsr_dcnto is a programmable word-length up counter with a dynamic count-to flag.

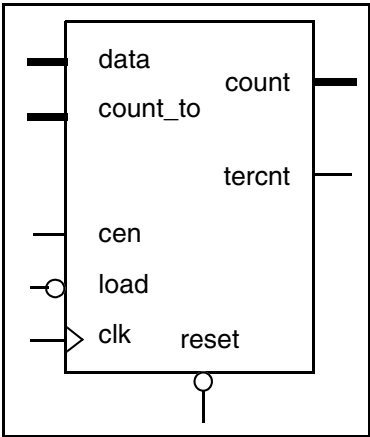


Table 1-1 Pin Description

Pin Name	Width	Direction	Function
data	<i>width</i> bits	Input	Input data
count_to	<i>width</i> bits	Input	Input count_to_bus
load	1 bit	Input	Input load data to counter, active low
cen	1 bit	Input	Input count enable
clk	1 bit	Input	Clock
reset	1 bit	Input	Asynchronous reset, active low
count	<i>width</i> bits	Output	Output count bus
tercnt	1 bit	Output	Output terminal count

Table 1-2 Parameter Description

Parameter	Legal Range ^a	Description
width	1 to 50	Word length of counter

a. The upper bound of the legal range is a guideline to ensure reasonable compile times

Table 1-3 Synthesis Implementations

Implementation Name	Function	License Feature Required
str	Synthesis model	DesignWare

Table 1-4 Simulation Models

Model	Function
DW03.DW03_LFSR_DCNTO_CFG_SIM	Design unit name for VHDL simulation
dw/dw03/src/DW03_lfsr_dcnto_sim.vhd	VHDL simulation model source code
dw/sim_ver/DW03_lfsr_dcnto.v	Verilog simulation model source code

Table 1-5 Counter Operation Truth Table

reset	load	cen	Operation
0	X	X	Reset
1	0	1	Load
1	X	0	Standby
1	1	1	Count up

DW03_lfsr_dcnto implements a counter as LFSR (linear feedback shift register), which also acts as a pseudo-random counter constructed as primitive characteristic polynomials.

The shift register is fed back from two or more taps. The number of taps does not increase with a large counter *width*.

A LFSR counter runs faster than a binary counter in synchronous systems because the first bit is calculated and the remaining bits are shifted.

DW03_lfsr_dcnto can be used for built-in test circuitry in VLSI chips and used as a modulus counter. For more information, see [Figure 1-1](#) on page 3.

Counter Function

The *width* is a generic parameter with an integer value ranging from 1 to 50.

The counter is loaded with data by asserting *load* (low) and applying data to *data*. The data load operation is synchronous with respect to the positive edge of *clk*.

The *count_to* is an input of pseudo-random binary sequences that ranges from *width* – 1 to 0. For a list of the primitive polynomials, see the [Logic-Sequential Overview](#).

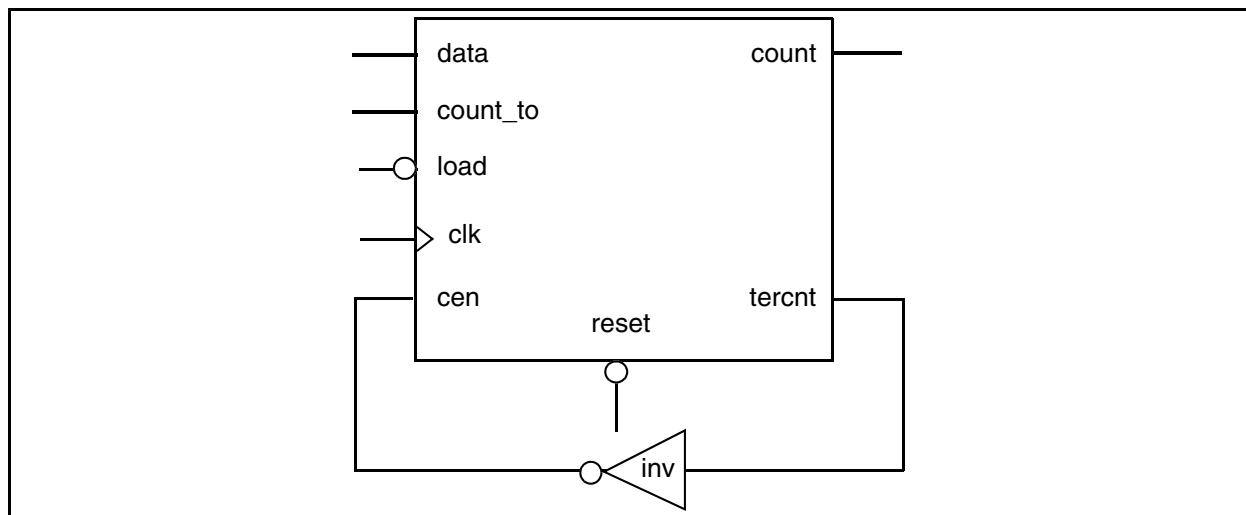
When the count enable pin, `cen`, is high, the counter is active. When `cen` is low, the counter is disabled, and count remains at the same value.

The `reset`, active low, is an asynchronous reset signal. When `reset` is low, the counter output is “00...00”. When `reset` is high, the counter operates normally.

The `count` is the output port of pseudo-random binary sequences, ranging from `width - 1` to 0. A value of $2^{width-2}$ (“11...11”) is an illegal state; therefore, the counter stops at “11...11”.

The `tercnt` is an output terminal count signal, active high. The `tercnt` output goes high for one clock cycle to indicate the different number of clock cycles between starting count to `count_to` value.

Figure 1-1 Counter Application: “count_to”



Timing Diagrams

Figure 1-2 and Figure 1-3 show various timing conditions for DW03_lfsr_dcnto.

Figure 1-2 Functional Operation: load and count_enable Sequence

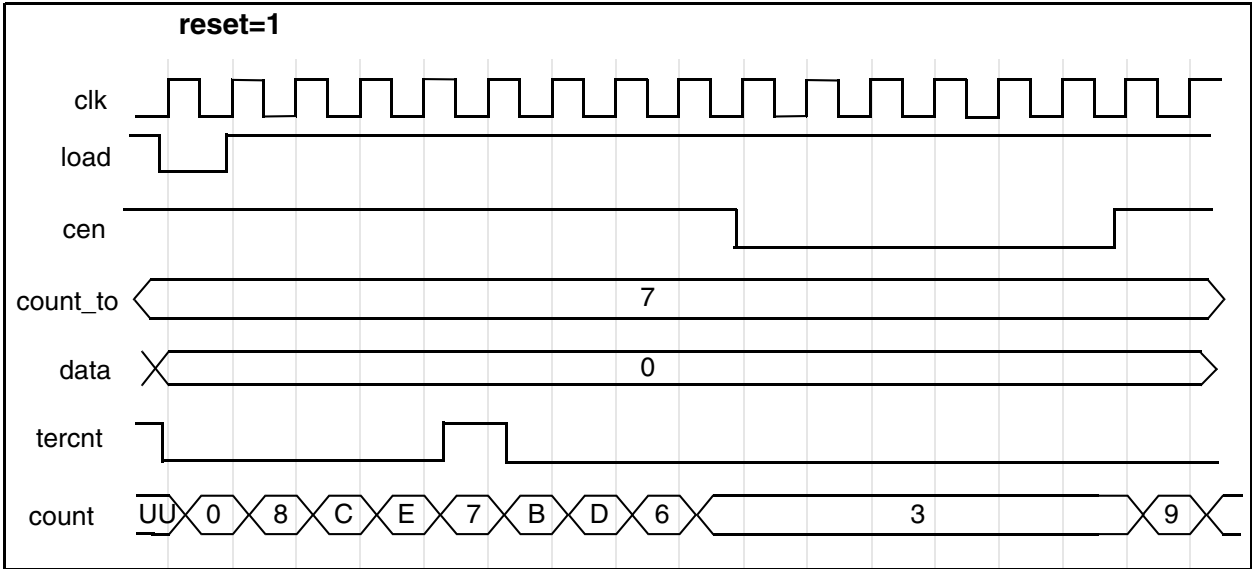
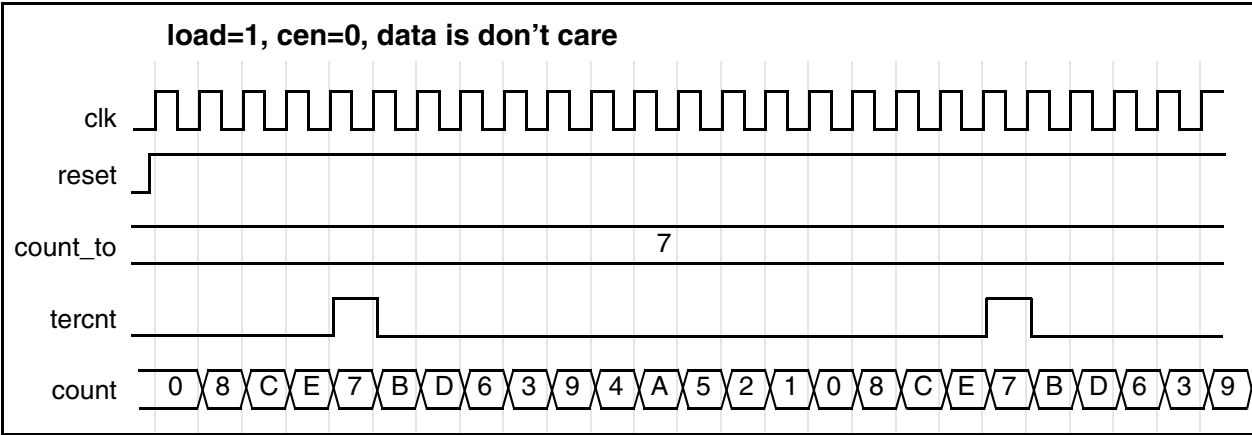


Figure 1-3 Functional Operation: reset and count_to Sequence



Related Topics

- [Logic – Sequential Overview](#)
- [DesignWare Building Block IP User Guide](#)

HDL Usage Through Component Instantiation - VHDL

```
library IEEE,DWARE;
use IEEE.std_logic_1164.all;
use DWARE.DWpackages.all;
use DWARE.DW_foundation_comp.all;

entity DW03_lfsr_dcnto_inst is
  generic (inst_width : INTEGER := 8);
  port (inst_data      : in std_logic_vector(inst_width-1 downto 0);
        inst_count_to  : in std_logic_vector(inst_width-1 downto 0);
        inst_load      : in std_logic;
        inst_cen       : in std_logic;
        inst_clk       : in std_logic;
        inst_reset     : in std_logic;
        count_inst     : out std_logic_vector(inst_width-1 downto 0);
        tercnt_inst    : out std_logic);
end DW03_lfsr_dcnto_inst;

architecture inst of DW03_lfsr_dcnto_inst is
begin

  -- Instance of DW03_lfsr_dcnto
  U1 : DW03_lfsr_dcnto
    generic map ( width => inst_width )
    port map ( data => inst_data, count_to => inst_count_to,
              load => inst_load, cen => inst_cen, clk => inst_clk,
              reset => inst_reset, count => count_inst,
              tercnt => tercnt_inst );

end inst;

-- pragma translate_off
configuration DW03_lfsr_dcnto_inst_cfg_inst of DW03_lfsr_dcnto_inst is
  for inst
    end for; -- inst
end DW03_lfsr_dcnto_inst_cfg_inst;
-- pragma translate_on
```

HDL Usage Through Component Instantiation - Verilog

```
module DW03_lfsr_dcnto_inst( inst_data, inst_count_to,
                             inst_load, inst_cen, inst_clk,
                             inst_reset, count_inst, tercnt_inst );

    parameter width = 8;

    input [width-1 : 0] inst_data;
    input [width-1 : 0] inst_count_to;
    input inst_load;
    input inst_cen;
    input inst_clk;
    input inst_reset;
    output [width-1 : 0] count_inst;
    output tercnt_inst;

    // Instance of DW03_lfsr_dcnto
    DW03_lfsr_dcnto #(width)
    U1 ( .data(inst_data), .count_to(inst_count_to),
        .load(inst_load), .cen(inst_cen), .clk(inst_clk),
        .reset(inst_reset), .count(count_inst), .tercnt(tercnt_inst) );

endmodule
```

Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

For a version of this datasheet with visible change bars, click [here](#).

Date	Release	Updates
September 2024	DWBB_202409.0	■ Updated date and version; no technical changes
December 2023	DWBB_202312.0	■ Updated date and version; no technical changes
March 2019	DWBB_201903.0	■ Removed minPower designation from this datasheet
January 2019	DWBB_201806.5	■ Updated example in “ HDL Usage Through Component Instantiation - VHDL ” on page 5 ■ Added this Revision History table and the document links on this page

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